UVA Health

The Role of Specialty Pharmacy in Coordinating and Monitoring Gene Therapy for Duchenne Muscular Dystrophy: The Elevidys® Experience at a Single Integrated Delivery Network Medical Center

Emily Chen, PharmD, BCPPS; Alisha Atchison, CPhT; Nathan Hart, PharmD; Angela Holian, PharmD, BCPS, MSCS; Joshua Weber, PharmD, MBA-HCM, CSP

University of Virginia Health, Charlottesville, Virginia

NASP NATIONAL ASSOCIATION OF SPECIALTY PHARMACY

BACKGROUND

- Duchenne muscular dystrophy (DMD) is a progressive genetic disorder caused by mutations in the dystrophin gene, leading to muscle dysfunction and premature mortality.
- Delandistrogene moxeparvovec-rokl (Elevidys®; DELA), delivers a functional copy of the dystrophin gene via an adeno-associated viral vector leading to microdystrophin expression, aiming to delay disease progression.
- DELA therapy carries significant off-target effects that require rigorous pre- and post-infusion monitoring.
- Specialty Pharmacy staff at UVA Health play a critical role in ensuring the safe and cost-effective administration of DELA throughout all stages of therapy.

OBJECTIVE

- Describe the effort and outcomes of specialty pharmacy staff involvement in the coordination execution, and monitoring of DELA therapy for pediatric patients with DMD at a single academic institution.

METHODS

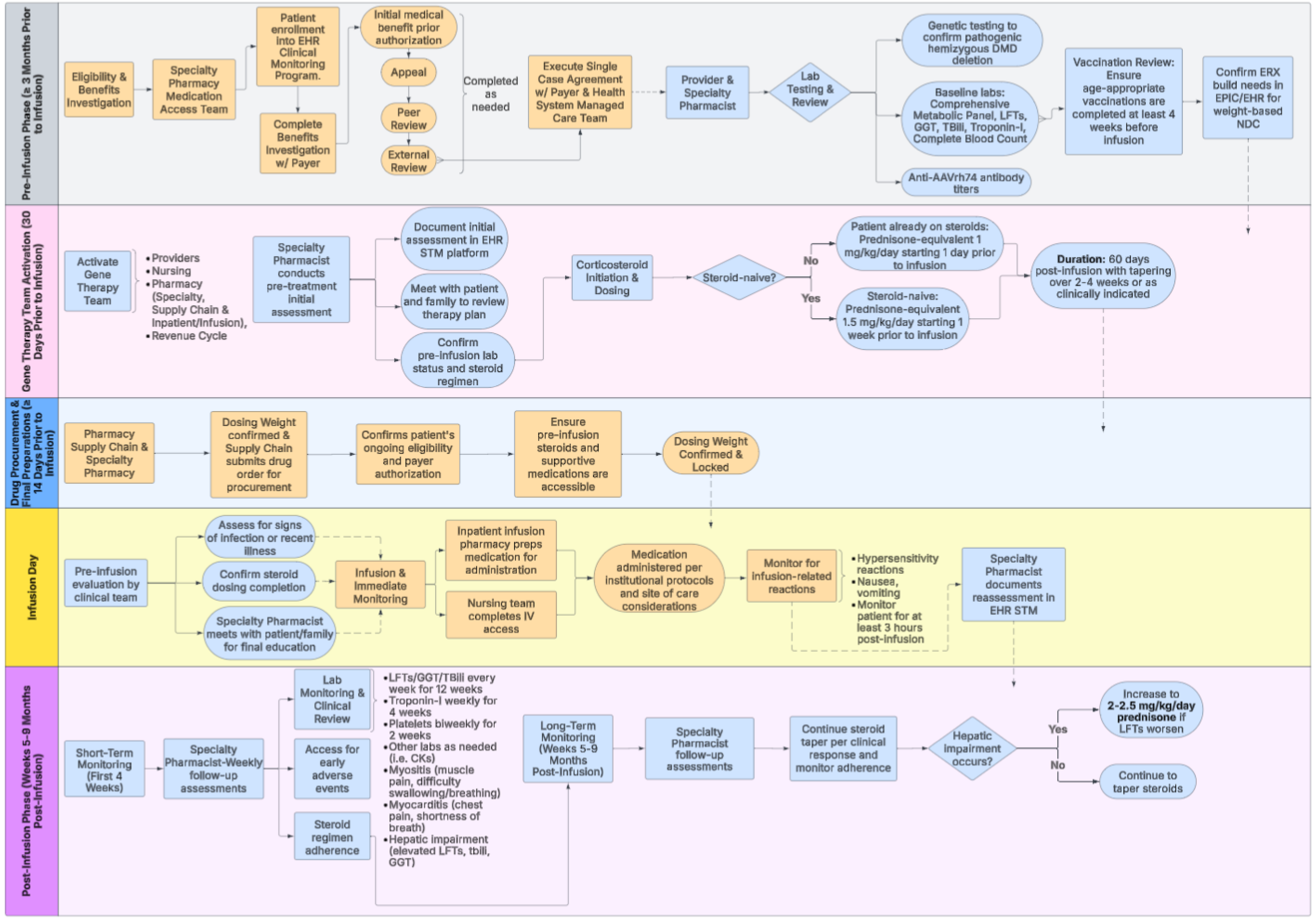
- Observational study conducted at UVA Health between March 1, 2024 and June 30, 2025, which included all patients who completed a DELA infusion.

RESULTS

Characteristic	Value
Total Patients	3
Median Age (range)	5 (4 to 12)
Median Dosing Weight, kg (range)	23 (21 to 45)
Ambulatory at baseline	3
Left ventricular ejection fraction at baseline	62% to 70%,
Baseline motor function *North Star Ambulatory Assessment	15 to 24/34
Median baseline LFTs (range) *Units reported in U/L	AST 189 (109 to 363) ALT 399 (216 to 519) GGT 15 (10 to 16)
Dystrophin gene exon deletions	48-50 45-50 49-50

RESULTS

Elevidys Gene Therapy Process Map for an IDN Academic Medical Center Health System



CK-Creatinine Kinase, EHR-Electronic Health Record, GGT-Gamma Glutamyl Transferase, ERX-Electronic Prescription Record, IV-Intravenous, LFT-Liver Function Tests, NDC-National Drug Code, STM-Specialty Therapy Management Program (e.g, Compass Rose)

This structured process ensures a seamless transition from pre-infusion preparation through post-infusion monitoring while aligning with payer requirements, clinical protocols, and patient safety considerations. This process map serves as a foundational workflow that is continuously refined in the evolving landscape of gene therapy

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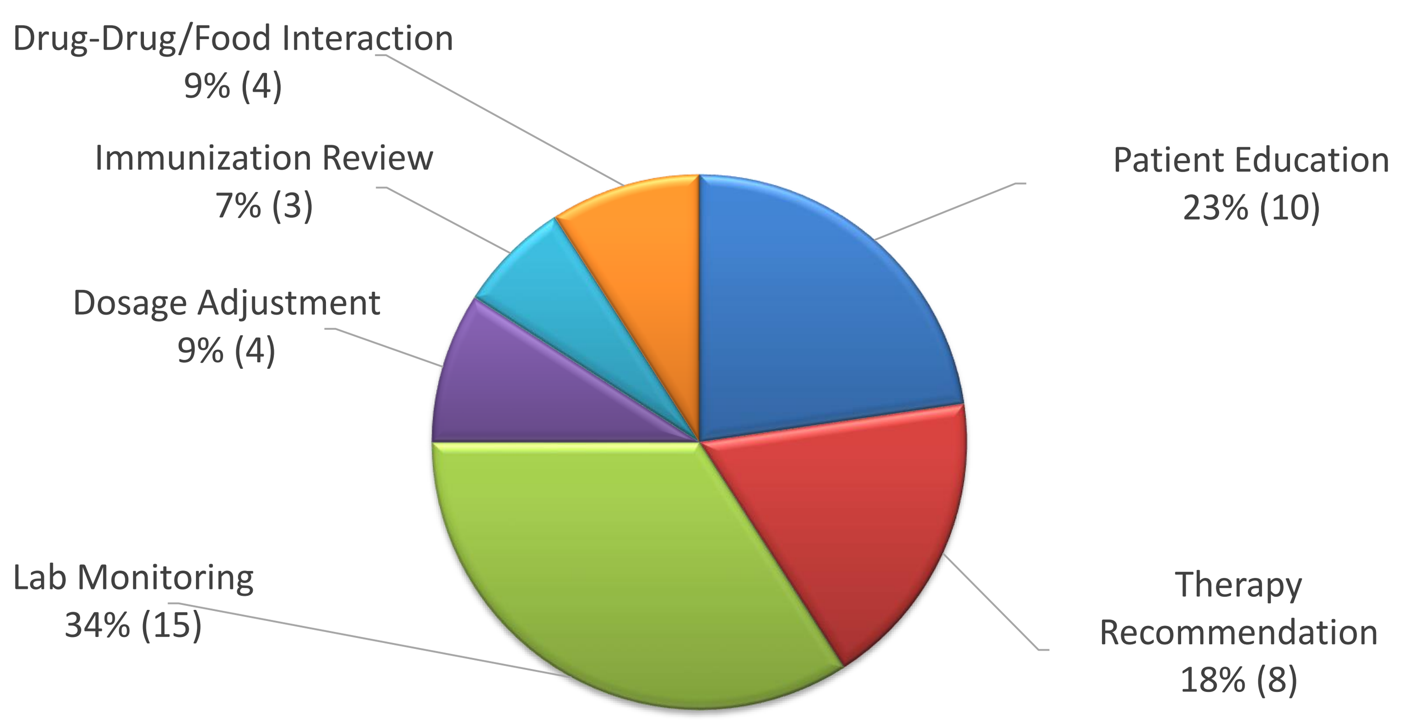
RESULTS

- Median time from prior authorization initiation to infusion administration is 128 days (53 to 214 days)
- Mean number of pharmacists visits per patient is 6.6 (5 to 7)
- Mean number of interventions per patient is 14 (12 to 17)
- No episodes of immune mediated myositis
- No episodes of acute liver failure
- All patients were on post-infusion prednisone for at least 60 days

Pharmacist Interventions by Type



CONCLUSIONS

- DELA therapy implementation necessitates intensive pharmacy involvement across all stages of treatment acquisition and management.
- This model may serve as a replicable framework for institutions implementing high-touch, high-risk therapies.

NEXT STEPS

- Recent sentinel events have prompted reconsideration of supportive care protocols.
- Ongoing research focuses on optimizing supportive care regimens to improve the safety of DELA infusions.

REFERENCES

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